

Grade Eight Science - Unit 2: Digestion and Nutrition

Final Assessment Information Sheet

Assessment Criteria:

• MYP Command Terms (Below) • Rubric: Criterion A - Knowing and Understanding (See Resources) • Rubric: Criterion D - Reflecting on the Impacts of Science (See Resources)
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MYP Command Terms that are used on this assessment:

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Present: Offer for display, observation, examination, or consideration	Suggest: Propose a solution, hypothesis, or other possible answer
Analyze: break down in order to bring out the essential elements or structure (To identify parts and relationships, and to interpret information to reach conclusions)	Summarize: Abstract a general theme or major points
Discuss: Offer a considered and balanced review that offers a range of arguments, factors, or hypothesis. Opinions or conclusions should be presented clearly and supported by appropriate evidence	Design: Produce a plan, simulation, or model
Organize: Put ideas and information into a proper or systematic order	Calculate: Obtain a numerical answer showing the relevant stages in the working

WHAT YOU HAVE TO DO

YOU & YOUR PARTNER ARE PROFESSIONAL NUTRITIONISTS

- You must give a presentation to the class about the work that you have done for one of your clients - What you have to present is described in Tasks 1 to 4 below - - Note that some parts of the work are collaborative, while others are individual Details about the presentation are in the box on the right	The presentation should: - Take no more than 7 minutes per student - this is the maximum time - Include a visual display (the type is up to you. Some examples are a Google Slides presentation or science fair project board, or your own idea) - Include all items outlined in the Tasks Table below must be included in the presentation - Include an MLA Works Cited for all research sources (minimum of four sources including the two provided for you in class - if you used them) Resources provided: 1. Micronutrients for Health: https://lpi.oregonstate.edu/sites/lpi.oregonstate.edu/files/pdf/mic/micronutrients_for_health.pdf 2. Atlantic Superstore Website: https://www.atlanticsuperstore.ca/en
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TASKS TABLE

TASK 1	Personalize your client: (Collaborative) - Use the information from the client ID card to help personalize your client - You can name your client - You can choose a specific city, town or region where your client will reside - You can add any more details that you would like the audience to know about your client (for example: is your client highly active or less active, does your client follow a specific religion or set of values that don't allow them to eat certain foods, what type of job do they do or are they retired?)	COLLABORATE: Partner 1 and Partner 2 work on this together
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TASK 2	Use your knowledge of nutrition, as well as your research skills, to design a healthy, nutritionally balanced meal plan for your client that contains <u>two</u> meals and <u>two</u> snacks. The meal plan should: • Contain two meals with the four types of macronutrients in each meal and two snacks with at least two different types of macronutrients. • Choose wisely: not all macronutrients are created equal. You must choose healthy options that are based on your clients individual needs. • Choose wisely: recall, your Design project focuses on ethical food. You must try to make ethical choices in your meal plan • Fall within the daily recommended calories for your client. You must research the proper calorie intake for your client based on the factors listed on their ID card, and you must calculate the total calories eaten per meal and snack per day. • You must give between 1 (minimum) and 3 (maximum) micronutrients that would benefit your client based on the factors given on the ID card. • Answer this debatable question: Does the client require nutritional supplements to meet their dietary needs?	COLLABORATE: • To find the total calories per day for the client • To decide who does which meal and snack • To decide what types of foods will be needed to meet the dietary requirements of the client • To decide what foods will allow us to make ethical choices • To brainstorm about the debatable question INDIVIDUAL: • Partner 1: designs one meal and one snack with calories • Partner 1: Answers the debatable question • Partner 2: designs one meal and one snack with calories • Partner 2: Answers the debatable question
TASK 3	Once your client's meal plan is in place, you must describe how some of the macronutrients in the diet are processed by the human digestive system Your description should include: • The physical changes that each macronutrient undergoes in the human digestive system • The chemical changes that each macronutrient undergoes in the digestive system • The proper use of digestive terminology (digestive organs and processes involved)	COLLABORATE: • To decide which macronutrient each partner will describe the digestion of • Each partner must choose a different macronutrient (carbohydrates, protein, or fat) to describe INDIVIDUAL: • Partner 1: Describes the physical and chemical digestion of one type of macronutrient in the meal plan • Partner 2: Describes the physical and chemical digestion of one type of macronutrient in the meal plan

TASK 4	Your client has a variety of dietary issues or other health problems In this section of the presentation each partner will: - Choose a dietary issue or other health problem that their client has - each partner must choose a different dietary issue or health problem Describe the ways that science is applied to address the specific problem or issue Discuss and analyse the implications of using science and its application to solve a specific problem or issue, interacting with one of the following factors: social, environmental, economic, cultural, or ethical	COLLABORATE: - To decide on which dietary restriction or health problem each partner will describe INDIVIDUAL: Partner 1: Describes the ways that science is applied to address the chosen problem or issue Discusses and analyses the implications of using science and its application to solve the chosen problem or issue, interacting with <u>one</u> of the following factors: social, environmental, economic, cultural, or ethical Partner 2:Describes the ways that science is applied to address the chosen problem or issue chosen Discusses and analyses the implications of using science and its application to solve the chosen problem or issue, interacting with <u>one</u> of the following factors: social, environmental, economic, cultural, or ethical
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RESOURCES

The presentation will be assessed using the following two MYP Criteria

- Rubric: Criterion A - Knowing and Understanding (Below) - Rubric: Criterion D - Reflecting on the Impacts of Science (Below) - MLA Citation Examples (see following pages) and MYP Command Terms (See Page 1)

Rubric 1: Criterion A: Knowing and Understanding

Achievement Level	Level Descriptor
0	The student does not reach a standard indicated by any of the descriptors below
1-2	The student is able to: recall scientific knowledge - apply scientific knowledge and understanding to suggest solutions to problems set in familiar situations apply information to make judgments
3-4	The student is able to: state scientific knowledge apply scientific knowledge and understanding to solve problems set in familiar situations apply information to make scientifically supported judgments.
5-6	The student is able to: outline scientific knowledge - apply scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations interpret information to make scientifically supported judgments
7-8	The student is able to: describe scientific knowledge apply scientific knowledge and understanding to solve problems set in familiar and unfamiliar situations analyze information to make scientifically supported judgments

Rubric 2: Criterion D: Reflecting on the Impacts of Science

Achievement Level	Level Descriptor
1-2	The student is able to: ● state the ways in which science is applied and used to address a specific problem or issue ● state the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● apply scientific language to communicate understanding but does so with limited success ● documents sources with limited success
3-4	The student is able to: ● outline the ways in which science is applied and used to address a specific problem or issue ● outline the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● sometimes apply scientific language to communicate understanding ● sometimes document sources completely
5-6	The student is able to: ● summarize the ways in which science is applied and used to address a specific problem or issue ● describe the implications of using science and its application to solve a specific problem or issue, interacting with a factor:social, environmental, economic, cultural, or ethical ● usually apply scientific language to communicate understanding clearly and precisely ● usually document sources completely
7-8	The student is able to: ● describe the ways in which science is applied and used to address a specific problem or issue ● discuss and analyse the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical ● consistently apply scientific language to communicate understanding clearly and precisely ● document sources completely

Feedback/Communication of Achievement

Descriptor	Emerging (Needs Improvement)		Developing (Satisfactory)		Proficient (Good)		Mastery (Excellent)	
Achievement Level (During Term)	1	2	3	4	5	6	7	8
Percentage (At Reporting Time)	0-39%	40%-49%	50%-62%	63%-75%	76%-82%	83%-89%	90%-95%	96%-100%

MLA CITATION EXAMPLES

Examples MLA Citations:

How to cite websites with a known author (one or multiple):

Format - one author

Author's Last Name, First Name. "Title of Webpage." *Title of Website*, Publisher, Date of Publication, URL.

Example - one author

Smith, John. "The Future of Artificial Intelligence." *Tech Innovations Today*, Tech Innovations, 15 Jan. 2025, www.techinnovationstoday.com/ai-future.

Format:

Author's Last Name, First Name, and Author's First Name Last Name. "Title of Webpage." *Title of Website*, Publisher, Date of Publication, URL.

Example - multiple authors

Lee, Sarah, and Michael Thomas. "The Impact of Renewable Energy on Global Economies." *Green Earth News*, Green Earth Media, 22 Feb. 2025, www.greenearthnews.com/renewable-energy-impact

How to cite websites with an unknown author:

Format - multiple authors

"Title of Webpage." *Title of Website*, Publisher, Date of Publication, URL.

Example - multiple authors:

"How to Stay Safe Online in 2025." *Digital Safety Guide*, Digital Safety Association, 3 Mar. 2025, www.digitalsafetyguide.com/online-safety-2025.

How to cite YouTube videos:

Format:

Author's Last Name, First Name (or Username). "Title of Video." *YouTube*, uploaded by Username, Date of Upload, URL.

Example:

Smith, John. "How AI is Transforming Healthcare." *YouTube*, uploaded by John Smith AI, 10 Feb. 2025, www.youtube.com/watch?v=example123